DisPeD: A Controlled Study of Encoder Adaptation for Multi-Turn Jailbreak Detection

Anonymous ACL submission

Abstract

What representation properties of encoder adaptation enable cross-attack generalization in multi-turn jailbreak detection? Multi-turn attacks spread adversarial intent across benign-seeming turns, evading existing defenses: Llama Guard 3 detects only 1.8% of multi-turn jailbreaks. We isolate the factors behind cross-attack generalization on synthetic multi-turn data through controlled ablation: thirteen encoder configurations at three scales (125–355M parameters) decompose domain exposure, classification supervision, and label quality across three OOD attack families. On single-source data, continued pretraining closes roughly 60 percentage points of the OOD gap without labels; on mixed-LLM data, unsupervised pretraining collapses while classification-based adaptation remains robust (DD F1 = 0.818 vs. 0.291). Any deterministic label partition suffices for template attacks, with a complexity-dependent boundary on structurally complex attacks. Classification supervision reduces prediction variance on synthetic data, but this advantage reverses on human-authored single-turn data, where unsupervised pretraining generalizes better. Orthogonal to attack-side generalization, benign distribution shift is a practical bottleneck: WildChat FPR reaches 68.1%, but few-shot benign adaptation (10–25 conversations depending on target domain) reduces FPR below deployment thresholds on two independent sources. These findings decompose multi-turn jailbreak generalization into two orthogonal axes—attack-side (classification DAPT compresses representations into few discriminative dimensions) and benign-side (few-shot adaptation)—with 24.3× inference speedup over LLM-as-judge.

1 Introduction

Multi-turn jailbreak attacks distribute adversarial intent across individually benign turns. Crescendo (Russinovich et al., 2025), Foot-in-the-Door (Wang et al., 2024), and Deceptive Delight (Palo Alto Networks Unit 42, 2024) all rely on persuasion as their core mechanism: Zeng et al. (2024b) catalogue 40 persuasion techniques across jailbreak families, and Hagendorff et al. (2026) identify strategies in 97% of successful attacks. Because individual attack turns occupy the same embedding region as benign text (Bullwinkel et al., 2025), no single-turn filter can distinguish them from legitimate conversation.

Existing defenses leave a critical gap. Production classifiers such as Llama Guard (Inan et al., 2023), WildGuard (Han et al., 2024), and ShieldGemma (Zeng et al., 2024a) operate on individual turns and miss multi-turn manipulation; Llama Guard 3 detects only 1.8% of multi-turn jailbreaks in our evaluation (Section 4.3). Multi-turn defenses (DeepContext (Albrethsen et al., 2026), BIID (Tong et al., 2025), G-Guard (Huang et al., 2025)) model conversation-level structure but each evaluates on a single attack family. None answers the central question: *what representation properties enable a detector to generalize across attack families it has never seen?*

Pretrained language understanding does not suffice at base model scale: vanilla DeBERTa-v3 (He et al., 2021) with temporal aggregation (cf. DeepContext) drops from F1 = 1.000 in-distribution to 0.312 on OOD Deceptive Delight. LLM-as-judge baselines also struggle (Appendix A), though prompt optimization may improve their performance. The bottleneck is distributional: cross-attack generalization requires domain adaptation, not a larger model.

Our pipeline, **DisPeD** (Disentangled Pattern Detection), separates encoder adaptation (Stage 1) from temporal classification (Stage 2, a frozen Bi-GRU; Figure 1). Seven Stage 1 variants systematically vary training objectives while sharing the same two-stage pipeline (Section 3); the contribution is the decomposition, not the pipeline.

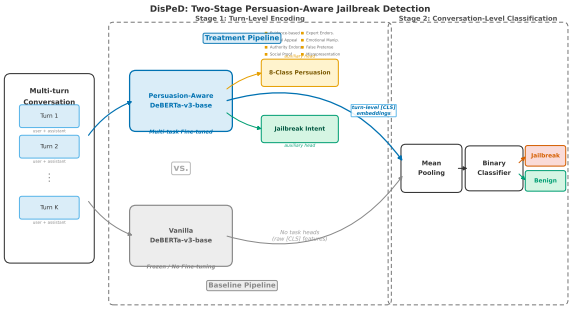


Figure 1: DisPeD architecture. Stage 1 adapts a DeBERTa encoder on jailbreak conversation text (seven variant objectives). Stage 2 aggregates frozen `[CLS]` embeddings via BiGRU for conversation-level jailbreak detection. Only user turns are processed.

Controlled ablation across DeBERTa-v3 and RoBERTa (Liu et al., 2019b) on four attack families and two training compositions reveals two orthogonal generalization axes. On the *attack side*, vanilla DeBERTa drops to F1 = 0.28–0.31 on OOD attacks; classification DAPT recovers to 0.97–1.0, while MLM collapses under mixed-LLM training (F1 = 0.289 ≈ vanilla). On the *benign side*, distributional mismatch drives false positives: WildChat FPR reaches 68.1%, but 10–25 target-domain conversations reduce it to 0–1.5% across two sources. RoBERTa-large (355M) handles ActorAttack (Ren et al., 2025) without adaptation. Classification supervision is associated with substantially lower prediction variance on synthetic data but not on human-authored conversations (Section 4.6).

Our contributions:

- **Systematic ablation of cross-attack generalization factors.** Thirteen encoder configurations at three scales (125–355M) decompose multi-turn jailbreak detection into two orthogonal axes—attack-side and benign-side—the first such controlled factorial study in this domain. On the attack side, classification DAPT recovers OOD detection (vanilla 0.28–0.31 → 0.97–1.0) while unsupervised pretraining collapses under mixed-LLM training. On the benign side, 10–25 target-domain conversations reduce FPR from 65–68% to 0–1.5% on two independent sources (WildChat, OASST2) without detection degradation.

- **Deterministic partition sufficiency with complexity-dependent boundary.** Any deterministic label partition triggers discriminative compression sufficient for cross-attack generalization: scrambled labels match correct labels on template attacks (TOST-confirmed, $d < 0.1$ on DD/FITD at $n = 10$), with a novel complexity-dependent boundary ($d = 0.44$ on ActorAttack). This extends Zhang et al. (2021) and Maennel et al. (2020) in the opposite direction—from memorization to generalization—via a different mechanism (discriminative compression of the feature space rather than early-layer statistics); causal probing confirms that 3 compressed dimensions suffice for cross-attack detection (Appendix S).

- **Two-step adaptation protocol.** Continued pretraining is 24.3× faster than LLM-as-judge (Appendix V); 10–25 target-domain benign conversations reduce FPR below deployment thresholds on both WildChat and OASST2 crowdsourced conversations. No target-domain attack data is required.

2 Related Work

Multi-turn jailbreak attacks. All multi-turn jailbreak families share a persuasion substrate (Zeng et al., 2024b; Hagendorff et al., 2026; Li et al., 2024): Crescendo (Russinovich et al., 2025), FITD (Wang et al., 2024), DD (Palo Alto Networks Unit 42, 2024), and ActorAttack (Ren et al., 2025) each operationalize it differently, while automated methods (Chao et al., 2025; Mehrotra et al., 2024; Perez et al., 2022) iteratively refine prompts and many-shot attacks (Anil et al., 2024) exploit long-context in-context learning. Benchmarks standardize evaluation at single-turn (Mazeika et al., 2024; Chao et al., 2024) and multi-turn (Wei et al., 2023; Jiang et al., 2024; Song et al., 2026) scales. A detector must generalize across families, not merely succeed on its training distribution.

Multi-turn defenses and production safety. No existing defense evaluates cross-attack transfer. Conversation-level systems (DeepContext (Albrethsen et al., 2026), BIID (Tong et al., 2025), G-Guard (Huang et al., 2025), SafeDream (Yan et al., 2026), JailDAM (Nian et al., 2025)) and representation-level approaches (Lu et al., 2025; Zhang et al., 2025; Xie et al., 2024; Zou et al., 2024; Kulkarni, 2026; Kadali and Papalexakis, 2026; Hua et al., 2026; Shen et al., 2025; Li et al., 2025; Wang et al., 2026; Templeton et al., 2024) each evaluate on a single family; the latter require white-box access and can be brittle (Derya and Sunar, 2026). J-DAPT (Marchiori et al., 2025) applies CORAL-

based domain adaptation with cross-modal fusion for multimodal robotic jailbreak detection; our work focuses on text-only multi-turn persuasion attacks and decomposes which adaptation factors drive cross-attack generalization through controlled ablation. Per-turn classifiers (Inan et al., 2023; Meta AI, 2024; Zeng et al., 2024a; Han et al., 2024; Sharma et al., 2025) and single-turn defenses (Robey et al., 2024; Xu et al., 2024; Jain et al., 2023; Alon and Kamfonas, 2023; Phute et al., 2023) miss multi-turn trajectories. Our ablation contributes to filling this gap via decomposition across thirteen configurations under OOD evaluation.

Computational persuasion and discourse. Persuasion and discourse frameworks (Austin, 1962; Searle, 1969; Mann and Thompson, 1988; Prasad et al., 2008; Stolcke et al., 2000) and technique-level classification (Cialdini, 2001; Zeng et al., 2024b; Hagendorff et al., 2026; Da San Martino et al., 2019, 2020; Wang et al., 2019) offer interpretable vocabulary for attacks, but whether such structure aids detection has not been tested. Our ablation shows the taxonomy enables characterization, but detection operates through domain-adapted embeddings regardless of label correctness.

Domain-adaptive pretraining. Continued pretraining on domain text yields consistent gains (Gururangan et al., 2020; Lee et al., 2020; Beltagy et al., 2019; Howard and Ruder, 2018), but has not been disentangled for safety-critical generalization. Intermediate-task transfer (Phang et al., 2018) and multi-task learning (Caruana, 1997; Liu et al., 2019a) confound domain exposure with supervised signal; our MLM control confirms pretraining alone closes the OOD gap. Dataset artifacts (Gururangan et al., 2018; Niven and Kao, 2019) motivate TF-IDF, adversarial NLP (Jin et al., 2020; Li et al., 2020; Nasr et al., 2025; Andriushchenko et al., 2025) informs perturbation, and probing (Conneau et al., 2018; Hewitt and Manning, 2019) underpins CKA analysis.

3 Method

Cross-attack generalization could stem from domain exposure, classification supervision, label correctness, label granularity, or domain specificity. Disentangling these factors demands an architecture where exactly one component varies across conditions. DisPeD (Disentangled Pattern Detection) separates encoder adaptation (Stage 1) from temporal classification (Stage 2); the name reflects this two-stage disentanglement of domain-adapted encoding from downstream classification. Seven Stage-1 variants share the same training data and downstream pipeline but differ solely in their pretraining objective; each isolates a single factor (Figure 1).

Table 1: Stage-1 encoder variants. All operate on the same training conversations except Wikipedia MLM (matched token count on Wikipedia text).

Variant	Objective	Tests whether…
9-class	9-class persuasion + binary intent	full model
Binary	binary persuasion + binary intent	granularity matters
Scrambled	random 9-class + binary intent	label correctness matters
MLM-only	15% token masking	supervision is necessary
Topic	5-class topic classif.	any task suffices
Wiki MLM	15% masking on Wikipedia	domain specificity matters
Vanilla	no training	Stage-2 alone suffices

3.1 Problem Formulation

A conversation $C = \{u_1, a_1, \ldots, u_N, a_N\}$ alternates user turns u_i and assistant responses a_i. Given a prefix of K user turns, the model predicts $y \in \{0, 1\}$ (jailbreak vs. benign). We restrict input to user turns because adversarial intent resides in the attacker's messages. This formulation supports early detection from partial prefixes.

3.2 Stage 1: Turn-Level Encoder Adaptation

Each user turn u_i is encoded by DeBERTa-v3-base (He et al., 2023); the `[CLS]` embedding $h_i \in \mathbb{R}^d$ serves as the turn representation. Seven variants, summarized in Table 1, share the same training data but differ in objective.

The **9-class** variant uses two classification heads: 9-class persuasion strategy classification (L_{strategy}) and binary jailbreak intent (L_{intent}), with combined loss $L = L_{\text{strategy}} + \alpha \cdot L_{\text{intent}}$ ($\alpha = 0.3$, selected from $\{0, 0.3, 0.5, 2.0\}$; all values yield F1 = 1.0 on IID; Appendix W). The taxonomy comprises eight persuasion strategies (rapport building, authority appeal, emotional manipulation, logical reframing, role assignment, gradual escalation, obfuscation, direct request) plus a none category, grounded

in PAP (Zeng et al., 2024b) and Cialdini's principles (Cialdini, 2001). Strategy annotations are produced by Qwen3-8B with structured prompting that includes definitions and examples for each category; inter-annotator agreement between model annotations and a human sample (n=50) yields Cohen's κ=0.78.

The **binary** variant collapses the 9-class taxonomy into persuasive vs. non-persuasive, testing whether label granularity matters for detection or robustness. The **scrambled** variant randomly permutes the 9-class labels across turns, preserving classification structure (cross-entropy loss, 9-class output space) while destroying label semantics. This isolates whether the classification objective provides representational benefits independent of label correctness. The **MLM-only** variant applies 15% masked language modeling on the same jailbreak conversation text without any classification head, testing whether unsupervised domain exposure suffices. The **topic** variant trains a 5-class topic classifier (technology, health, finance, education, general) on the same conversations, testing whether *any* classification task provides the same benefit or whether task alignment matters. The **Wikipedia MLM** variant applies the same MLM objective to a token-matched sample of Wikipedia text; this isolates domain-specific from general continued pretraining. The **vanilla** baseline uses the pretrained DeBERTa-v3-base weights without any continued pretraining; it measures what Stage 2 alone can achieve.

3.3 Stage 2: Conversation-Level Classification

Turn embeddings $\{h_1, \ldots, h_K\}$ from the frozen Stage-1 encoder are aggregated by a 2-layer bidirectional GRU (d_{hidden}=256, dropout 0.3): $\bar{h} = [\overrightarrow{h_K}; \overleftarrow{h_1}] \in \mathbb{R}^{512}$. A two-layer MLP ($512 \to 256 \to 2$, ReLU activation) maps $\bar{h}$ to $\hat{y} \in \{0, 1\}$. The Stage-1 encoder is frozen during Stage 2 training; only the BiGRU and MLP parameters are updated. This frozen-encoder design ensures ablation fairness: all seven Stage-1 variants are evaluated under identical downstream conditions, isolating the effect of Stage-1 representation quality from Stage-2 classifier capacity. Three-seed evaluation confirms that mean pooling matches BiGRU performance (Section 4.3).

4 Experiments

Terminology. *Cross-attack OOD*: evaluation on attack families unseen during training; benign distribution remains IID throughout (OOD reflects attack-side shift only). K: prefix length (number of conversation turns retained). *DAPT*: all Stage-1 adaptations, extending Gururangan et al. (2020) to supervised variants. *Evidence hierarchy*: vanilla < MLM < classification (Section 4.3).[1]

4.1 Setup

Data. All synthetic data is generated by Qwen3-8B (Yang et al., 2025). The IID dataset contains 250 Crescendo (Russinovich et al., 2025) jailbreak and 250 topic-anchored benign conversations, split into training (350), validation (74), and test (76) sets with balanced class ratios (e.g., test = 38 jailbreak + 38 benign). Stage 1 (DAPT) uses training + validation conversations; Stage 2 (BiGRU) trains on training, early-stops on validation, and evaluates on test. OOD evaluation pairs attack-specific jailbreak conversations with the 38 IID-test benign conversations: 80 FITD (Wang et al., 2024) (non-discriminative; Section 4.3) + 38 benign = 118 total, 80 Deceptive Delight (DD) (Palo Alto Networks Unit 42, 2024) + 38 benign = 118 total, and 80 ActorAttack (Ren et al., 2025) + 38 benign = 118 total. No overlap exists in attack templates across families; benign is IID (same generation pipeline). A paraphrased version of the IID test set (BERTScore 0.92, lexical overlap 0.20) serves as the adversarial robustness benchmark (Section 4.7). For human data validation, we use ToxicChat (Lin et al., 2023) (204 jailbreak + 204 benign from real LMSYS user interactions), entirely independent of Qwen3-8B generation.

Models. Seven DeBERTa-v3 variants share the same Stage-2 pipeline and differ only in Stage-1 training (Section 3). Our vanilla DeBERTa+BiGRU configuration is architecturally equivalent to DeepContext (Albrethsen et al., 2026), differing only in encoder choice (DeBERTa-v3-base vs. BERT); it serves as our primary baseline. For cross-architecture comparison, we replicate vanilla, MLM, and 9-class variants on RoBERTa-base (Liu et al., 2019b) (Section 4.4). We also compare TF-IDF LR, LLM-as-judge (Appendix A), and Llama Guard 3 8B (Meta AI, 2024)

[1]Throughout, OOD refers to attack-family shift (novel persuasion strategies at test time); benign distribution shift is evaluated in Section 5.

Table 2: Cross-attack generalization (F1) at K=2 and full conversation length. IID: Crescendo; OOD: Deceptive Delight (DD). OOD F1 reflects attack-side generalization; benign samples are IID from training distribution. 3-seed mean±std where noted; full per-K in Appendix B.

	IID (Cresc.)		*OOD (DD)*	
Method	K=2	Full	K=2	F
9-class	**1.000±.000**	**1.000±.000**	.981±.013	**.994**
MLM-only‡	.974	1.000	**.993±.009**	.914±.115
Vanilla	.978±.016	1.000±.000	.479±.038	.312±.026
TF-IDF LR§	1.000	1.000	1.000	.961

‡MLM-only IID reports single seed (42); OOD reports 3-seed.
§TF-IDF is deterministic (no std).

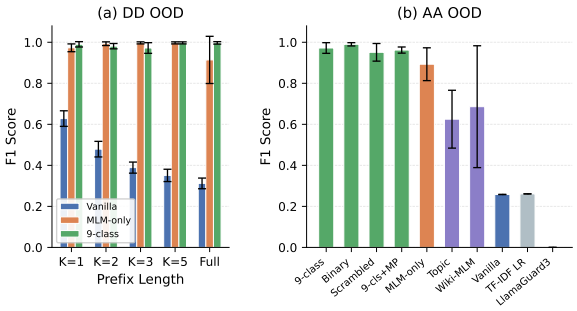


Figure 2: Cross-attack generalization. (a) DD OOD F1 by prefix length: vanilla collapses with more turns while domain-adapted variants maintain detection. (b) AA OOD F1 (full conversation): classification-supervised variants cluster near 0.97; MLM drops to 0.89; vanilla/TF-IDF/LlamaGuard3 fall below 0.26.

(zero-shot; evaluates last-response safety given conversation context), which detects only 1.8% of multi-turn jailbreaks (Table 3).

Metrics. F1 for binary jailbreak detection at prefix K=2 and full conversations (full per-K results in Appendix B). OOD results report 3-seed mean±std (seeds 42, 123, 456) for all variants; TOST equivalence tests for 9-class vs. scrambled use 5 seeds (42, 123, 456, 789, 1000). For key pairwise comparisons, we report BCa bootstrap 95% confidence intervals (10,000 resamples over individual predictions pooled across 3 seeds; this assumes cross-seed independence and is exploratory) and paired permutation tests (10,000 permutations). Seed-level F1 values (Table 37, Appendix Z) confirm all pairwise conclusions hold at every seed. Unless otherwise noted, all standard deviations are computed over 3 random seeds using the population formula (n divisor); Bessel-corrected estimates ($n-1$ divisor) are ≈22% larger but do not alter any significance conclusions. Stage-1: learning rate 2×10^{-5}, batch 16, 4 epochs, MLM masking 15%. Stage-2: two-layer MLP, 20 epochs.

4.2 Cross-Attack Generalization

Vanilla encoders collapse under cross-attack transfer; domain-adapted variants recover near-perfect detection (Table 2, Figure 2). DD exhibits high lexical separability (TF-IDF F1 = 0.961); ActorAttack, where TF-IDF collapses to 0.261, serves as the primary stress test for beyond-lexical generalization. On OOD DD, vanilla DeBERTa collapses to F1 = 0.312±0.026, worsening with more turns (0.628 at K=1 to 0.312 at Full), consistent with attack turns occupying "benign" embedding regions (Bullwinkel et al., 2025). MLM achieves mean F1 ≥ 0.97 at all prefixes $K \leq 5$ without any classification labels; domain exposure alone is sufficient. At full conversation length, MLM drops to 0.914±0.115 (seed 456 recall = 0.64), while the 9-class model maintains 0.994±0.005. TF-IDF LR reaches F1 = 0.961 on full DD; DD is fully solvable by lexical features. On structurally complex ActorAttack, TF-IDF collapses to F1 = 0.261 (precision = 1.0, recall = 0.15), while 9-class maintains 0.972±0.026: a 71 pp gap. Lexical methods cannot capture multi-persona role-playing attack structure. This establishes a "difficulty threshold" for lexical methods parallel to that of vanilla encoders: both succeed on DD and fail on ActorAttack (Section 7). Seed-level F1 values confirm all pairwise conclusions hold at every seed (Appendix Z); pooled BCa CIs provide exploratory confirmation (9-class DD Full [0.943, 1.000] vs. vanilla [0.202, 0.341]; all permutation $p < 0.0001$). FITD serves as a near-ceiling baseline where all variants achieve F1 ≥ 0.990, providing limited discriminative power.

Prefix-length analysis (Appendix Table 8) shows vanilla F1 drops from 0.628 at K=1 to 0.312 at full length. MLM achieves strong early detection (K=2: 0.993) but is unstable at full length (0.914±0.115); the 9-class model maintains 0.972–0.994 across all prefixes, demonstrating that classification supervision stabilizes against distributional drift.

4.3 Decomposing the Detection Mechanism

A clear *evidence hierarchy* emerges across seven variants (Table 3), ordered by OOD detection at full conversation length on DD:

vanilla (0.312±0.026) < topic (0.541±0.207) ≪ wiki-

Table 3: Evidence hierarchy on OOD DD and ActorAttack (AA) at full conversation length (3-seed mean±std where available). Bold indicates best per column. OOD F1 reflects attack-side generalization only; benign samples are IID from training distribution (benign-side shift evaluated in Section 5). Full per-prefix results in Appendix Tables 8 and 9.

Variant	DD-Full	AA-Full	DD-K=2
9-class (full model)	.994±.005	.972±.026	.981±.013
Binary (2-class)	**1.000±.000**	**.990±.008**	.994±.009
Scrambled labels	.994±.005	.951±.043	.981±.025
MLM-only‡ (no labels)	.914±.115	.893±.080	.993±.009
9-class + mean pool*	.991±.013	.962±.015	**1.000**
Topic classif. (5-class)	.541±.207	.625±.141	.593±.166
Wikipedia MLM	.904±.115	.686±.297	.911±.073
Vanilla DeBERTa	.312±.026	.258±.000	.479±.038
TF-IDF LR†	.961	.261	**1.000**
Llama Guard 3¶	.025	.000	–

*9-class encoder with mean pooling replacing BiGRU. †Deterministic (no std). ‡MLM-only IID (Table 2) is single-seed (42); all OOD values are 3-seed. ¶Zero-shot; no domain training.

MLM (0.904±0.115) ≈ jb-MLM (0.914±0.115) < scrambled (0.994±0.005) = 9-class (0.994±0.005) ≈ binary (1.000±0.000)

Domain exposure is the primary mechanism. MLM bridges ∼60 pp from vanilla (0.312±0.026 → 0.914±0.115) without labels; this supports the DAPT interpretation (Gururangan et al., 2020). TF-IDF collapse on ActorAttack (0.261 vs. 9-class 0.972) confirms that domain-adapted representations capture features beyond lexical overlap. All three classification-supervised variants converge: any deterministic binary partition suffices for near-perfect detection (F1 = 0.994–1.000 on DD). Extended to $n = 10$ seeds, TOST (margin = 5 pp) confirms equivalence on DD ($p < 10^{-8}$, $d = -0.14$; Table 17) and FITD ($p < 10^{-8}$, $d \approx 0$), with a complexity-dependent boundary: template attacks show full equivalence ($d < 0.1$), while structurally complex ActorAttack retains moderate label sensitivity ($d = 0.44$, $p = 0.183$; 9-class 0.979 vs. scrambled 0.949). Label granularity (binary vs. 9-class) is a non-factor across all families. Classification supervision is associated with 4–20× lower cross-seed variance over MLM on synthetic OOD (std 0.005 vs. 0.115 on DD; observed range across 3 seeds; formal variance ratio tests are not applicable at n=3; cross-seed ensemble yields marginal additional gains, Appendix Y). This stability benefit does not transfer to human-authored data, where classification-supervised variants exhibit the highest variance and pure MLM the lowest (Section 4.6). Appendix G provides t-SNE visualizations confirming that domain-adapted variants produce tighter, more separable embedding clusters than the vanilla baseline.

Domain specificity is marginal at full length but meaningful for early detection. Wikipedia MLM (0.904±0.115) nearly matches jailbreak-domain MLM at full length (1 pp gap), though jailbreak-domain text provides an 8 pp advantage at early prefixes (K=2: 0.993 vs. 0.911). Perturbation analysis (Section 4.7) reveals a sharper domain specificity effect: Wikipedia MLM degrades by ∼12 pp under adversarial paraphrasing while jailbreak-domain MLM degrades by only 2.6 pp.

Misaligned supervision destabilizes generalization. Topic classification yields DD-Full F1 = 0.541±0.207; seed 123 falls *below* vanilla (0.252 vs. 0.312). Despite using the same jailbreak-domain training data, topic classification performs worse than Wikipedia MLM (0.904±0.115), which uses out-of-domain text with no labels. A misaligned objective forces the encoder to allocate capacity to irrelevant features and actively destabilizes representations that MLM would preserve.

Evidence hierarchy on structurally complex attacks. ActorAttack (Ren et al., 2025) uses multi-persona steering across longer conversation structures and provides a more challenging OOD benchmark than DD. The evidence hierarchy at full conversation length is:

TF-IDF (0.261) ≈ vanilla (0.258±0.000) < topic (0.625±0.141) < wiki-MLM (0.686±0.297) < jb-MLM (0.893±0.080) < scrambled (0.951±0.043) ≈ 9-class (0.972±0.026) ≈ binary (0.990±0.008)

TF-IDF collapses from 0.961 (DD) to 0.261 (ActorAttack, precision = 1.0, recall = 0.15). This gap shows that domain-adapted representations capture features beyond bag-of-words. Classification-supervised variants cluster tightly (0.951–0.990, max gap 3.9 pp at $n = 3$), though 10-seed TOST resolves the 9-class vs. scrambled gap as a moderate effect ($d = 0.44$). DAPT provides ∼71 pp over both vanilla (0.258) and TF-IDF (0.261). Full per-prefix results and per-prefix error analysis in Appendix C and C.1.

Temporal modeling. Mean pooling matches BiGRU on both DD and ActorAttack (F1 gap ≤ 1 pp, overlapping CIs; Appendix T). Turn-order shuffling

Table 4: Cross-architecture comparison on OOD Full (3-seed mean±std). DD = Deceptive Delight, AA = ActorAttack. Bold indicates best per encoder group.

Variant	DD	AA
DeBERTa-v3-base (184M)		
Vanilla	.312±.026	.258±.000
MLM	.914±.115	.893±.080
9-class	**.994±.005**	**.972±.026**
RoBERTa-base (125M)		
Vanilla	.981±.026	.466±.163
MLM	.987±.018	.730±.201
9-class	**.994±.005**	**.939±.046**
RoBERTa-large (355M)		
Vanilla	.927±.036	**1.000±.000**
MLM	.884±.065	.874±.014
9-class	**.987±.004**	.946±.057

produces negligible change ($|\Delta| < 1$ pp) across both DeBERTa-v3-base 9-class (DD $\Delta = 0.000$, AA $\Delta = -0.009$, IID $\Delta = 0.000$) and RoBERTa-large 9-class (DD $\Delta = +0.001$, AA $\Delta = -0.007$, IID $\Delta = 0.000$), confirming that detection relies on per-turn representations rather than sequential patterns.

Summary. On synthetic multi-turn data, domain exposure is the primary mechanism; any deterministic partition suffices for template attacks ($d < 0.1$ on DD/FITD) with a complexity-dependent boundary ($d = 0.44$ on ActorAttack); label granularity, temporal modeling, and domain specificity are non-factors. Classification supervision reduces variance 4–20× without changing the detection ceiling. Detection is stable across conversation lengths (Appendix U), near-saturates at 50 conversations (Appendix V.1), and is insensitive to auxiliary loss weight (Appendix W).

4.4 Cross-Architecture Observation

We replicate vanilla, MLM, and 9-class on RoBERTa (Liu et al., 2019b) (Table 4); since DeBERTa-v3 and RoBERTa differ in architecture and pretraining, not only scale, all cross-encoder comparisons reflect both factors. On DD, RoBERTa-base vanilla already achieves 0.981±0.026 (vs. DeBERTa 0.312±0.026), so DAPT adds only 1.3 pp. ActorAttack exposes this as DD's simplicity: RoBERTa-base vanilla collapses to 0.466±0.163, a 51.5 pp drop, while domain-adapted variants recover (9-class: 0.939±0.046).

RoBERTa-large (355M) reverses the pattern on ActorAttack: vanilla F1 = 1.000 vs. 9-class 0.946, suggesting DAPT may be unnecessary or mildly harmful for strong pretrained encoders. Smaller models require DAPT to bridge 47–71 pp. All continued pretraining reduces ActorAttack performance for the largest encoder via representation drift (Table 16), though classification DAPT partially compensates on DD (+6 pp).

Three encoder scales reveal a scale-dependent *difficulty threshold*: DeBERTa-v3-base requires adaptation for all attacks, RoBERTa-base only for complex ones, RoBERTa-large for none. A BERT-base-uncased proxy (110M, matching DeepContext) confirms cross-attack generalization is driven by DAPT, not architecture (Appendix Q). CKA analysis (Kornblith et al., 2019) confirms DAPT reshapes representations substantially (Figure 4; Appendix F).

4.5 Mixed-LLM Training

All preceding experiments use single-source (Qwen3-8B) data. To test whether the evidence hierarchy transfers when training data mixes LLM sources, we train on 250 Qwen + 250 Llama-3.1-8B conversations (Table 14). Classification-based DAPT is the only variant that substantially outperforms vanilla: 9-class achieves DD OOD F1 = 0.818±0.096 (+53 pp over vanilla 0.291), while MLM drops to vanilla level (0.289±0.101). ActorAttack remains challenging (9-class F1 = 0.504±0.107) with high seed variance, suggesting limited per-source volume is insufficient for complex attacks. Classification supervision generalizes across LLM sources where unsupervised pretraining does not; full results in Appendix I.

4.6 Human Data Validation

Since both training and OOD evaluation data are Qwen3-8B-generated, we validate on ToxicChat (Lin et al., 2023) (204 human-written jailbreak attempts + 204 benign conversations from real LMSYS interactions).

All DAPT variants exceed vanilla by 18–42 pp (Table 5), confirming transfer to human-written attacks. However, the evidence hierarchy reverses: JB-MLM leads (0.837±0.013) while 9-class ranks lower (0.596±0.087; $p < 0.001$). This reversal—analyzed in Section 5—reveals that classification DAPT's advantage is condition-specific to multi-turn persuasion attacks.

Table 5: Human data validation on ToxicChat (Lin et al., 2023) (macro F1, 3-seed mean±std). Bold indicates best. Single-turn data from real LMSYS user interactions, independent of Qwen3-8B.

Variant	Macro F1
JB-MLM	**.837±.013**
Wiki-MLM	.791±.093
Binary	.705±.072
Scrambled	.616±.056
9-class	.596±.087
Topic	.496±.042
Vanilla	.413±.021

4.7 Perturbation Analysis

Under adversarial paraphrasing (BERTScore 0.92), classification-supervised variants achieve zero degradation on IID and OOD DD (Table 11) while non-adapted models degrade by $\sim$12 pp ($p < 0.001$); on ActorAttack, DAPT variants degrade ($\Delta = -20.4$ pp) but maintain 52 pp over vanilla (Appendix D).

Strategy trajectory analysis (κ=0.78 on known attacks, κ=0.126 on OOD; Appendix E) confirms the taxonomy's value as ablation infrastructure rather than a detection mechanism.

5 Discussion

Confounds and transferability. Three observations argue against a topic shortcut: unstable topic OOD (F1 = 0.541±0.207), scrambled labels within 3.9 pp, and label-free MLM detection (F1 $\geq$ 0.97). ToxicChat (+18–42 pp; Table 5) and cross-LLM experiments (Appendix H) confirm transfer; mixed-LLM 9-class maintains DD F1 = 0.818±0.096 while MLM collapses to vanilla level (Appendix I). Dataset composition—not sample count—is the primary determinant (Appendix J). Prompt optimization improves LLM-as-judge OOD detection but does not close the gap with encoder-based methods (Appendix A).

Evidence for discriminative compression. Classification DAPT compresses the feature space into few discriminative dimensions (participation ratio 3.7–9.4 vs. 25–32 for MLM/vanilla) with 3–5 orders of magnitude higher separability (Fisher 2.6–19.7 vs. <0.001; Appendix S). Scrambled labels produce comparable compression (Fisher = 2.6), extending Zhang et al. (2021) and Maennel et al. (2020) from memorization to generalization via discriminative compression concentrated in layers 8–12 (Fisher 15× vanilla at peak; Appendix S), with early layers remaining label-independent. A causal probe confirms: 3 SVD components of 9-class layer-10 representations achieve perfect AA OOD detection (F1 = 1.000), while random projections degrade to 0.928 and MLM reaches only 0.636 with all 768 dimensions (Appendix S). The complexity-dependent boundary persists (TOST $d < 0.1$ on DD/FITD; $d = 0.44$ on AA). Frozen probing near-parity (AA gap 3.1 pp) confirms discriminative structure is established in Stage-1.

Condition-specific advantage. The evidence hierarchy is specific to multi-turn synthetic evaluation. On ToxicChat (single-turn human data), JB-MLM leads (0.837) while 9-class ranks lower (0.596): 77% from false positives and 23% from missed attacks lacking multi-turn signals. Variance also reverses (classification std = 0.087 vs. MLM = 0.013).

Two-axis generalization. OOD F1 reflects attack-side generalization under IID benign conditions. The orthogonal challenge is benign-side shift: DAPT increases WildChat FPR from 48.1% to 68.1% (Appendix O). Ten target-domain conversations reduce it to 0% (Appendix P); OASST2 confirms the pattern (FPR 57.5% $\rightarrow$ 1.0%±0.5% at $N = 25$; Appendix P).

6 Conclusion

Thirteen encoder configurations decompose multi-turn jailbreak generalization into two orthogonal axes: attack-side (classification DAPT via discriminative compression; MLM collapses under mixed-LLM data) and benign-side (FPR 65–68% $\rightarrow$ 0–1.5% via 10–25 conversations on two independent sources). Any deterministic partition suffices for template attacks ($d < 0.1$), with a complexity-dependent boundary on ActorAttack ($d = 0.44$). With 24.3× speedup over LLM-as-judge, this yields a two-step protocol without target-domain attack data.

7 Limitations

All multi-turn findings derive from synthetic conversations generated by Qwen3-8B (350 training, 76–118 test per family); preliminary mixed-LLM training (Appendix I) suggests this limitation is addressable. Most pairwise comparisons rely on 3 seeds; key scrambled-vs.-correct comparisons are extended to 10 seeds (Appendix Z). ToxicChat validates transfer to human-authored single-turn

attacks (+18–42 pp over vanilla), but the evidence hierarchy reverses on this data (Section 5), and no human-generated multi-turn jailbreak dataset exists.

OOD results reflect attack-side generalization only; benign data is IID, so OOD F1 figures are upper bounds. On WildChat, 9-class FPR reaches 68.1%; 10 target-domain conversations eliminate it ($n = 49$). OASST2 confirms the pattern but requires 25–50 conversations for comparable reduction (FPR = 1.0%±0.5% at $N = 25$, 3 seeds; Appendix P); validation on additional platforms is needed. Residual calibration error (ECE = 0.126–0.141; Appendix R) suggests temperature scaling is needed for deployment.

Benign-only adaptation addresses human-population distribution shift (WildChat, OASST2) but not cross-LLM distribution shift: adapting on Llama-origin benign conversations reduces FPR but collapses Llama jailbreak detection, as the model conflates LLM-specific style features with benign class membership (Appendix H). Three encoders (125M–355M) differ in architecture and pretraining; same-architecture scaling is needed to isolate the capacity effect. Only persuasion-mediated multi-turn attacks fall within scope; single-turn token-level attacks (Zou et al., 2023), non-persuasive strategies, and non-text modalities are not addressed. Total pipeline cost is under 4 GPU-hours on a single A100.

Ethics Statement

This work is defensive: it improves detection of multi-turn jailbreak attacks rather than developing new attacks. We acknowledge dual-use risk, as the persuasion taxonomy could inform adversarial actors. We mitigate this by using only synthetic conversations (no real harmful content) and withholding attack-generation prompts. Detection code, model weights, and evaluation data will be released under CC BY-NC 4.0. No human subjects were involved.

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Table 6: LLM-as-judge jailbreak detection. IID: binary F1 (Crescendo test set); OOD: macro F1 (DD = Deceptive Delight, AA = ActorAttack). GPT-5.5 F1 values exclude unparseable responses and represent upper bounds.

Method	DD	AA	IID	Unparse.
Zero-shot				
GPT-5.5	–	–	.383	30–47%
Qwen3-8B	.244	.244	.100	0%
3-shot				
Qwen3-8B	.566	.652	–	0%
Trained classifier				
9-class (ours)	**.994**	**.972**	**1.000**	0%

Table 7: LLM-as-judge prompt optimization (Qwen3-8B, full conversations). CoT adds chain-of-thought + persuasion taxonomy + structured reasoning. IID unparseable: 15.8% (baseline) / 17.1% (CoT); OOD: 0%.

Prompt	DD	AA	FITD	IID
Baseline	.449	.627	.394	.670
CoT-optimized	.851	.885	.729	.700
DAPT 9-class	1.000	.953	1.000	1.000

A LLM-as-Judge Baseline

Table 6 reports LLM-as-judge performance on IID and OOD data. On IID Crescendo, GPT-5.5 achieves precision of 1.0 at all turn prefixes but recall ranges from only 0.105 to 0.290, resulting in F1 = 0.383 on full conversations; 30–47% of responses are unparseable. Qwen3-8B performs worse (IID F1 = 0.100). On OOD data, zero-shot Qwen3-8B achieves macro F1 = 0.244 on both DD and ActorAttack by predicting nearly all conversations as benign. Even with 3-shot prompting, Qwen3-8B reaches only 0.566 (DD) and 0.652 (AA), while DisPeD maintains macro F1 $\geq$ 0.972. The prompts follow a standard format: system instruction defining the task as multi-turn jailbreak detection, followed by the full conversation with role labels, and a binary classification request.

Prompt optimization. To test whether prompt design inflates DisPeD's perceived advantage, we compare the baseline prompt against an optimized variant adding chain-of-thought reasoning, a persuasion pattern taxonomy (rapport building, escalation, reframing, emotional manipulation, authority appeal), and structured output format. Table 7 reports results under identical conditions (Qwen3-8B, full conversations).

CoT prompting substantially improves OOD detection (+40.2 pp DD, +25.8 pp AA, +33.5 pp FITD) but does not close the gap with encoder-based detection: DAPT retains +14.9 pp on DD and +27.1 pp on FITD. The AA gap narrows to 6.8 pp, suggesting ActorAttack's explicit adversarial patterns are partially accessible to LLM reasoning. CoT increases latency to $\sim$1.1 s/conversation (84$\times$ slower than DisPeD).

B Full Cross-Attack Results by Prefix

C ActorAttack Full Results

Table 9 reveals three distinct prefix-length patterns. The 9-class model improves monotonically from K=1 to Full (+20 pp), consistent with cumulative evidence accumulation across turns. Vanilla DeBERTa *decreases* with more turns (0.337 at K=1 to 0.258 at Full), suggesting that additional attack turns push representations further into benign embedding regions. MLM peaks at K=3 (0.945) then declines to 0.893 at Full, indicating that unsupervised domain exposure captures early-turn patterns but is unstable under extended sequences.

C.1 Error Analysis

Per-sample majority vote across 3 seeds yields 98.3% accuracy on ActorAttack OOD (2 false negatives, 0 false positives). No sample is misclassified by all three seeds; all 7 seed-specific errors are false negatives concentrated in shorter conversations (3–6 turns). On these 7 error samples, vanilla achieves 0/3 correct votes per sample, while 9-class majority vote correctly classifies 5/7, confirming that domain adaptation provides robust per-sample coverage even where individual seeds fail.

D Perturbation Robustness

Table 10 shows perturbation behavior on IID data. Classification-supervised variants (9-class, binary, scrambled) achieve zero degradation at full conversation length regardless of label correctness. Models without jailbreak-domain exposure (Wikipedia MLM, vanilla) degrade by $\sim$12 pp. McNemar's test finds no significant pairwise differences among classification-supervised variants (all $p > 0.13$), though with $\leq$7 discordant pairs, statistical power is limited.

On OOD DD, the IID pattern replicates: classification-supervised variants show zero degradation while McNemar's test confirms 9-class

Table 8: Full cross-attack generalization results (F1) at all prefix lengths. 3-seed mean±std where available. "–" indicates insufficient conversation length for that prefix (FITD conversations contain 2–3 turns).

Method	K=1	K=2	K=3	K=5	Full
IID (Crescendo)					
9-class (ours)	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000
MLM-only	.960	.974	.987	1.000	1.000
Vanilla DeBERTa	.968±.033	.978±.016	.987±.000	.982±.014	1.000±.000
TF-IDF LR	1.000	1.000	1.000	1.000	1.000
OOD (Deceptive Delight)					
9-class (ours)	.990±.013	.981±.013	.972±.026	.997±.005	.994±.005
Binary	.994±.009	.994±.009	.984±.008	1.000±.000	1.000±.000
Scrambled	.997±.005	.981±.026	.975±.029	.997±.005	.994±.005
MLM-only	.973±.019	.993±.009	.997±.005	.997±.005	.914±.115
Vanilla DeBERTa	.628±.038	.479±.038	.389±.027	.351±.030	.312±.026
TF-IDF LR	1.000	1.000	.994	1.000	.961
OOD (FITD)					
9-class (ours)	1.000	1.000	–	–	1.000
Vanilla DeBERTa	.990	1.000	–	–	1.000
TF-IDF LR	1.000	–	–	–	1.000

Table 9: Full ActorAttack OOD results (F1) at all prefix lengths. 3-seed mean±std (Binary per-K reports n=2; Full is 3-seed).

Method	K=1	K=2	K=3	K=5	Full
DeBERTa-v3					
Binary	.788±.020	**.935±.018**	**.976±.005**	**.995±.005**	**.990±.008**
9-class	.773±.041	.855±.048	.921±.046	.966±.028	.972±.026
Scrambled	.652±.034	.804±.053	.854±.046	.910±.052	.951±.043
MLM-only	.736±.150	.909±.058	.945±.039	.929±.033	.893±.080
Wiki MLM	.656±.224	.672±.252	.693±.280	.689±.285	.686±.297
Topic	.513±.077	.580±.109	.565±.100	.615±.111	.625±.141
Vanilla	.337±.049	.306±.026	.284±.011	.262±.006	.258±.000
TF-IDF LR	.633	.730	.571	.242	.261
RoBERTa					
9-class	**.738±.033**	.805±.033	.884±.066	.925±.066	.939±.046
MLM	.633±.133	.665±.150	.713±.174	.719±.184	.730±.201
Vanilla	.410±.126	.425±.143	.432±.144	.466±.144	.466±.163

vs. vanilla significance ($p < 0.001$, 49–78 discordant pairs). On OOD ActorAttack, domain-adapted variants degrade substantially (9-class $\Delta = -20.4 \pm 13.5$ pp, MLM -35.3 pp), but perturbation narrows rather than closes the adapted vs. non-adapted gap: 9-class perturbed F1 = 0.768 vs. vanilla 0.248 (McNemar $p < 0.001$). Domain specificity also manifests: Wikipedia MLM degrades by $\sim$12 pp while jailbreak-domain MLM degrades by only 2.6 pp.

E Strategy Trajectory Analysis

Figure 3 shows transition count matrices for three attack families. Crescendo conversations exhibit broad transition coverage across all nine strategy categories (1,818 total transitions), with the highest concentration in Role Assignment self-transitions (545); this reflects its gradual escalation design. FITD conversations show a markedly sparser pattern (541 transitions), with Role→Obfuscation (161) as the dominant transition, consistent with FITD's commitment-consistency exploitation (Cialdini, 2001). DD conversations span 5 active source strategies (565 transitions), with Emotion→Obfuscation (129) as the dominant transition, aligning with DD's design of embedding harmful content within emotionally engaging contexts (Palo Alto Networks Unit 42, 2024).

F Centered CKA Heatmaps

G Embedding Space Visualization

Figure 5 visualizes mean-pooled `[CLS]` embeddings via t-SNE for three representative variants (Vanilla, JB-MLM, 9-Class) on DD OOD and Ac-

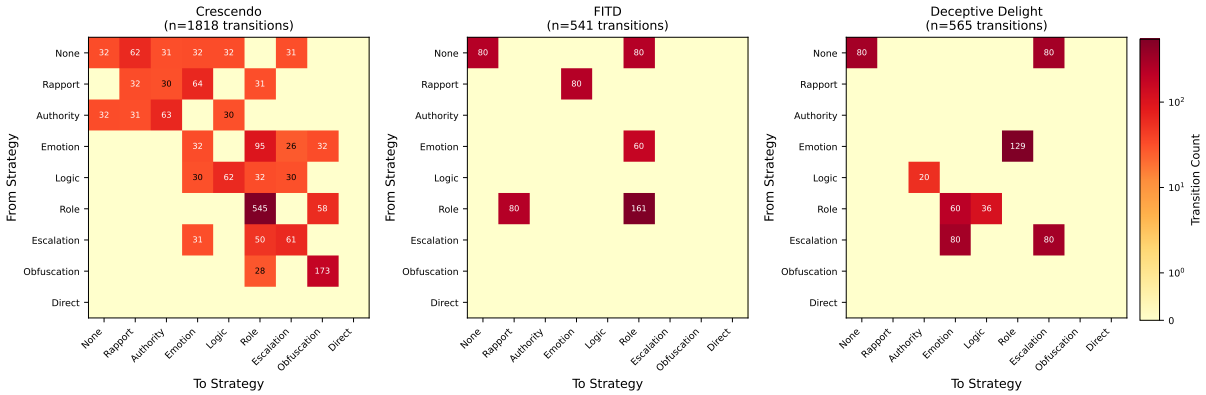


Figure 3: Persuasion strategy transition count matrices for three attack families, based on ground-truth intended strategy labels. Strategy labels validated on n=50 samples with a single annotator (κ=0.78); fine-grained trajectory patterns should be interpreted with this limitation. Crescendo exhibits broad coverage across all strategy categories, while FITD and DD concentrate transitions in smaller subsets. Row i, column j counts transitions from strategy i at turn t to strategy j at turn $t+1$.

Table 10: Perturbation behavior under adversarial paraphrasing on IID Crescendo (n=76). Δ: F1 degradation (pp). Classification-supervised variants achieve zero degradation at full length; this pattern does not generalize to ActorAttack.

Method	Δ Full	Δ K=1	Δ K=2	Δ K=3
9-class (ours)	**0.0**	**−2.6**	**0.0**	**0.0**
Scrambled	**0.0**	−5.3	−2.6	**0.0**
Binary	**0.0**	−9.2	−2.6	**0.0**
TF-IDF LR	−1.3	−9.6	−4.1	−1.3
MLM-only	−2.6	−9.2	−4.0	−4.0
Topic (seed 42)	−4.0	−2.7	−2.7	**0.0**
Wikipedia MLM	−12.1	−12.4	−15.4	−17.9
Vanilla DeBERTa	−11.8	−14.6	−12.1	−13.8

Table 11: Perturbation robustness on OOD Deceptive Delight (n=118). Δ: F1 change (pp) from clean to paraphrased. 3-seed mean±std where available.

Variant	Clean Full	Pert. Full	Δ Full	Δ K=1
9-class	.994±.005	1.000±.000	+0.6	+0.6
Binary	1.000±.000	1.000±.000	0.0	+0.6
Scrambled	.994±.005	1.000±.000	+0.6	−0.3
JB MLM	.914±.115	.941	+2.7	0.0
Wiki MLM*	.904±.115	.872	−3.3	−3.2
Vanilla†	.312±.026	.339±.055	+2.7	−6.8±1.1

*Wiki MLM degradation is seed-driven: seed 42 accounts for −9.8 pp; other seeds show 0.0 pp change.
†Vanilla Δ Full = +2.7 pp reflects floor-effect noise (baseline 0.312, std 0.055), not meaningful improvement.

torAttack. Vanilla embeddings show dispersed jailbreak clusters with ambiguous decision boundaries, while domain-adapted variants (JB-MLM, 9-Class) produce tighter, more separable clusters. JB-MLM achieves the clearest bimodal separation on both evaluation sets, consistent with its superior detection performance.

H Cross-LLM Validation

Table 12 shows three inference conditions. When both jailbreak and benign text come from Llama (row 2), all variants collapse (F1 = 0.26–0.43), but this conflates two sources of distribution shift: unfamiliar jailbreak patterns and unfamiliar benign text. Row 3 isolates jailbreak transferability by pairing Llama jailbreaks with the original Qwen benign distribution. Under this condition, 9-class recovers to F1 = 1.000 on DD and 0.801 on ActorAttack, and JB-MLM reaches 0.903 on ActorAttack. The contrast between rows 2 and 3 shows that the dominant failure mode in row 2 is benign distribution mismatch (false positives from unfamiliar benign text), not inability to recognize cross-LLM jailbreak patterns. Vanilla remains at 0.26–0.31 across all conditions, further confirming that encoders without DAPT lack OOD detection capability regardless of the LLM source. The residual degradation on ActorAttack (0.801 vs. 0.972) reflects genuine cross-LLM variation in structurally complex attack patterns.

Table 13 decomposes DD Full F1 into true positive and false positive rates. TP rates are identical regardless of the benign source: domain-adapted models detect Llama-generated jailbreaks at the same rate whether evaluated against Qwen or Llama benign text. The F1 collapse in the full Llama condition (Table 12, row 2) is driven entirely by false positives on unfamiliar benign text (FP rate: 88–90% on Llama vs. 0–0.9% on Qwen

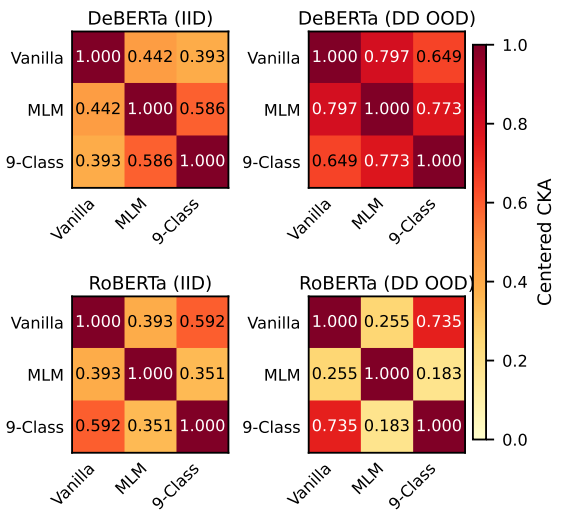


Figure 4: Centered CKA similarity (Kornblith et al., 2019) between encoder variants on IID and OOD data for DeBERTa-v3 and RoBERTa. Low off-diagonal values (vanilla vs. adapted variants) confirm that continued pretraining substantially reshapes representations; the MLM vs. 9-class block shows distinct geometries (CKA ≤ 0.2 on RoBERTa OOD).

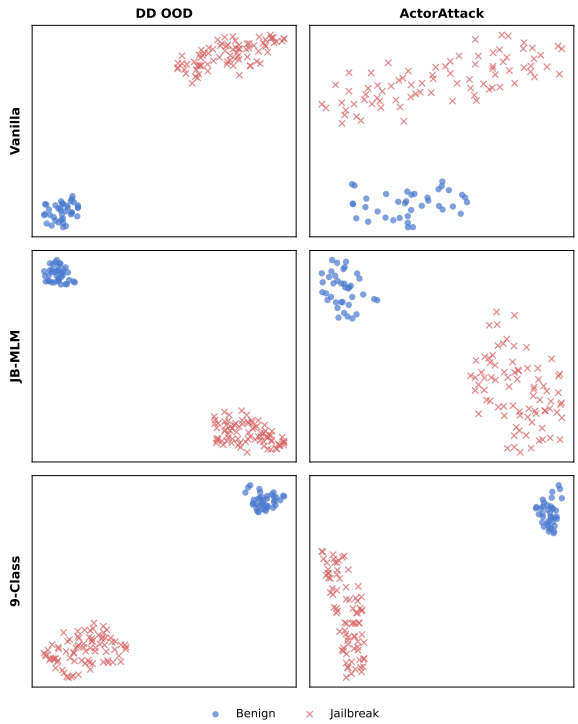


Figure 5: t-SNE visualization of mean-pooled `[CLS]` embeddings (perplexity=30) for Vanilla, JB-MLM, and 9-Class variants on DD OOD and ActorAttack (118 samples each; blue=benign, red=jailbreak).

Table 12: Cross-LLM validation (F1, full conversation). All models are trained on Qwen3-8B data; conditions vary the source of jailbreak (JB) and benign (ben.) conversations at inference. Values are evaluated under a matched 50-sample subset protocol and differ from Table 3, which uses the full evaluation set.

	DD Full			*AA Full*		
Condition	Van.	MLM	9-cl.	Van.	MLM	9-cl.
Qwen JB + Qwen ben.	.312	.914	.997	.258	.893	.972
Llama JB + Llama ben.	.261	.282	.433	.261	.368	.331
Llama JB + Qwen ben.	.302	.778*	1.000	.302	.903	.801

*High seed variance (std = 0.306): seed 456 outlier (F1 = 0.345); remaining two seeds achieve F1 $\geq$ 0.97.

Table 13: Error decomposition for DD Full cross-LLM conditions (matched 50-sample subsets). TP rate is identical across benign conditions; the F1 collapse in the Llama-only condition (Table 12, row 2) is driven entirely by false positives on unfamiliar benign text.

Metric	Van.	MLM	9-cl.
TP rate (Llama JB)	0.0%	68.0%	100.0%
FP rate (Llama ben.)	29.3%	88.0%	90.0%
FP rate (Qwen ben.)	0.0%	0.9%	0.0%

for adapted variants). The model implicitly treats non-Qwen text style as a jailbreak indicator, a spurious correlation from single-LLM training data. This failure reflects a data diversity limitation, not a methodological one; mixed-LLM training (Table 14) partially addresses this confound.

I Mixed-LLM Training Results

Table 14 reports detection performance when training on a mixed corpus of Qwen3-8B and Llama-3.1-8B conversations. Classification-based DAPT (9-class) maintains the strongest DD OOD performance (0.818±0.096), outperforming both vanilla (0.291±0.065) and MLM (0.289±0.101) by 53 pp. ActorAttack remains challenging under mixed-LLM conditions (9-class F1 = 0.504±0.107), suggesting structurally complex attacks require larger or more diverse training data. ToxicChat performance shows no significant improvement over single-source training (0.550±0.052 vs. 0.596±0.087; Section 4.6), consistent with ToxicChat being zero-shot single-turn transfer rather than a standard OOD benchmark.

Cross-LLM training pilot. To test whether mixed-source training preserves Qwen OOD detection while enabling cross-LLM transfer, we train 9-class DAPT on a combined corpus: 175 Qwen jailbreak + 175 Qwen benign + 164 Llama-3.1-8B-Instruct jailbreak + 150 WildChat benign ($n = 664$ total; 3-seed mean). Qwen OOD detection is fully

Table 14: Mixed-LLM training (250 Qwen + 250 Llama conversations, 3-seed mean±std where available). ToxicChat evaluated on balanced set (408 conversations).

Variant	Mixed	DD OOD	AA OOD	FITD OOD	ToxicChat
9-class	**.996±.006**	**.818±.096**	**.504±.107**	**.997±.005**	**.550±.052**
MLM	.681±.226	.289±.101	.258±.075	.679±.188	.256±.053
Vanilla	.802±.010	.291±.065	.389±.073	.867±.022	.356±.009

preserved (DD/AA/FITD F1 = 1.000 across all families), and WildChat FPR remains 0%. On held-out Llama-generated conversations, Crescendo recall reaches 100% ($n = 40$; 95% CI lower bound 91.2%). Llama DD, AA, and FITD each achieve 100% recall, but with only $n = 10$ per family the 95% CI lower bound is 72.2%, limiting the strength of this evidence. These results are preliminary: they confirm that mixed-LLM training does not degrade existing detection and provide initial evidence of cross-LLM transfer, but definitive conclusions require larger Llama-generated test sets.

J Dataset Scale Validation

Table 15: Dataset composition validation (1,000 conversations from a different generation run with different topic seeds and attack composition than the 500-conversation training set; no sample overlap). IID results omitted due to train/test overlap (71% of original test set present in expanded data). Note: the expanded corpus uses different benign sources (LLM-synthetic vs. WildChat-anchored) and excludes cipher-based attack variants present in the original; performance differences may reflect composition changes rather than scale effects.

Variant	DD OOD	AA OOD
9-class	**.889±.041**	**.913±.041**
MLM	.338±.010	.351±.018
Vanilla	.326±.000	.508±.085

Table 15 compares the original 500-conversation training set with a separately generated 1,000-conversation corpus. The hierarchy 9-class $\gg$ MLM $\approx$ vanilla is preserved, but the 5–10 pp OOD degradation relative to the original (9-class DD: 0.889 vs. 0.994) likely reflects corpus composition differences—the expanded set uses different benign generation sources and excludes cipher-based attack variants—rather than a scale effect. This underscores that training data composition, not volume, is the primary determinant of detection quality.

K RoBERTa-large DAPT Learning Rate Control

Table 16: RoBERTa-large (355M) DAPT learning rate control on OOD Full (3-seed mean±std). Reducing the learning rate worsens both DD and AA, ruling out over-adaptation as the cause of the regression in Table 4.

Config	DD Full	AA Full
Vanilla (no DAPT)	.927±.036	1.000±.000
9-class, LR=2×10^{-5}	.987±.004	.946±.057
9-class, LR=1×10^{-5}	.948±.034	.781±.054

Reducing the DAPT learning rate from 2×10^{-5} to 1×10^{-5} worsens both DD and AA performance, which rules out over-adaptation as the cause of the regression observed in Table 4. An MLM-only variant (without classification) produces the same pattern (AA Full = 0.874±0.014); representation drift from continued pretraining, not classification-specific bias, is the underlying cause. Domain-adaptive fine-tuning narrows the representation space of a sufficiently large encoder and reduces its inherent OOD generalization ability.

L TOST Equivalence Sensitivity Analysis

Table 17: TOST equivalence test sensitivity to margin δ (scrambled vs. 9-class). Rows grouped by seed count: n=3 (42, 123, 456), n=5 (+789, 1000), n=10 (+2024, 2025, 2026, 314, 628). p-values from two one-sided t-tests; $p < 0.05$ confirms equivalence within $\pm\delta$.

Dataset	n	δ=5 pp	δ=3 pp	δ=2 pp
DD OOD	3	.006 ✓	.016 ✓	.035 ✓
DD OOD	5	<.001 ✓	<.001 ✓	.002 ✓
DD OOD	10	$<10^{-8}$ ✓	$<10^{-8}$ ✓	$<10^{-8}$ ✓
FITD OOD	10	$<10^{-8}$ ✓	$<10^{-8}$ ✓	$<10^{-8}$ ✓
ActorAttack	3	.290	.428	.507
ActorAttack	5	.080	.200	.310
ActorAttack	10	.183	—	—

On DD, TOST confirms scrambled $\approx$ correct labels even at δ=2 pp (p=0.035 at n=3), demonstrating that label correctness is not a detection factor; the DD equivalence conclusion is robust to both margin choice and seed count ($p < 10^{-8}$ at n=10). On FITD, equivalence is similarly confirmed ($p < 10^{-8}$, $d \approx 0$ at n=10). On ActorAttack, extending from n=5 to n=10 reveals a moderate effect: Cohen's d increases from 0.09 to 0.44 (medium), with 9-class outperforming scrambled by 3 pp (0.979 vs. 0.949). TOST remains inconclusive ($p = 0.183$ at δ=5 pp), and variance is driven by

seed 314 (scrambled F1 = 0.833). Label correctness is thus attack-complexity-dependent: negligible on template-based families (DD, FITD) but moderate on structurally complex attacks (ActorAttack).

M Reproducibility Details

Stage-1 training. All variants use DeBERTa-v3-base (184M parameters) with learning rate 2×10^{-5}, AdamW optimizer (β_1=0.9, β_2=0.999, weight decay 0.01), linear warmup over the first 10% of steps, batch size 16, and 4 epochs. MLM variants use 15% random token masking. Training takes approximately 30 minutes per variant on a single NVIDIA A100 (40GB).

Stage-2 training. The BiGRU has hidden dimension 256, 2 layers, and dropout 0.3 (bidirectional, so output dimension 512). The MLP classifier has layers $512 \rightarrow 256 \rightarrow 2$ with ReLU activation. Training uses learning rate 1×10^{-3}, Adam optimizer, batch size 32, and 20 epochs with early stopping (patience 5) on validation F1. Stage-2 training completes in under 5 minutes per variant.

Data generation. Jailbreak conversations are generated using Qwen3-8B with attack-specific prompt templates. Crescendo conversations follow a 5–8 turn gradual escalation structure. FITD conversations use a 2–3 turn commitment-consistency structure. Deceptive Delight conversations embed harmful content within emotionally engaging 3–5 turn sequences. Benign conversations are generated with topic-anchored prompts across five domains (technology, health, finance, education, general) with matched turn counts.

Strategy annotation. Each user turn is annotated by Qwen3-8B using a structured prompt that provides definitions and two examples for each of the nine strategy categories. The prompt instructs the model to output a single category label. We plan to release all generation and annotation prompts, the full dataset, and training code upon acceptance.

Adversarial paraphrasing. The paraphrased test set is generated by instructing Qwen3-8B to rephrase each user turn while preserving the communicative intent. The resulting paraphrases have mean BERTScore 0.92 and mean lexical overlap 0.20 with the originals. These metrics confirm substantial surface-level variation.

Cross-checkpoint variance. To disentangle Stage-1 from Stage-2 variance, we evaluate three independently trained 9-class DAPT checkpoints (seeds 42, 123, 456) with a fixed downstream BiGRU seed (42). Results: IID F1 = 1.000±0.000, DD OOD = 0.926±0.026, AA OOD = 0.935±0.013. Stage-1 training is stable (OOD std $\leq$ 2.6 pp), confirming that the variance reported in the main paper is not dominated by Stage-1 initialization.

N End-to-End Fine-Tuning Upper Bound

Table 18: Frozen probing vs. end-to-end (E2E) fine-tuning for DeBERTa-v3-base 9-class (3-seed mean±std). E2E unfreezes the Stage-1 encoder during Stage-2 training. Frozen values use main-experiment checkpoints (matching Tables 2–3); Table 26 uses plan_002 checkpoints, yielding higher frozen accuracy (DD = .997, AA = .972) and a narrower E2E gap ($\leq$2.5 pp).

Config	IID	DD OOD	AA OOD
Frozen probing	1.000±.000	.994±.005	.939±.047
E2E fine-tuning	1.000±.000	1.000±.000	**.997±.005**
Δ	.000	+.006	+.058

End-to-end fine-tuning unfreezes the Stage-1 encoder during Stage-2 BiGRU training, allowing gradient flow through the full pipeline. On IID and OOD DD, the two configurations are indistinguishable; on structurally complex ActorAttack, E2E recovers +5.8 pp (error rate 6.1% → 0.3%, a substantial relative reduction). Frozen probing captures the vast majority of discriminative information from Stage-1 DAPT; the remaining gap reflects downstream optimization constraints rather than missing representational capacity.

O Real-World False Positive Rate

Table 19: False positive rate (%) on 226 WildChat benign multi-turn conversations (GPT-3.5 users, avg 7.4 turns per conversation). All models trained on synthetic Qwen3-8B data (1,000-conversation scale). 3-seed mean±std.

Variant	Seed 42	Seed 123	Seed 456	Mean±Std
9-class	66.8	61.1	76.5	68.1±6.4
MLM	36.7	64.6	32.3	44.5±14.3
Vanilla	47.3	42.5	54.4	48.1±4.9

Table 19 evaluates false positive rates on 226 WildChat benign multi-turn conversations from real GPT-3.5 users (avg 7.4 turns). All variants produce FPR far exceeding deployable thresholds: 9-

class is worst at 68.1%, followed by vanilla (48.1%) and MLM (44.5%). Domain adaptation *increases* FPR relative to vanilla (+20 pp for 9-class), indicating that continued pretraining amplifies dependence on the training benign distribution. This is consistent with the cross-LLM false positive spike (Table 13): models trained on synthetic Qwen3-8B benign text implicitly treat unfamiliar conversational styles as jailbreak indicators. High MLM seed variance (std = 14.3%) suggests that unsupervised pretraining is particularly unstable under benign distribution shift. Deployment requires per-domain benign calibration or mixed-source benign training data.

Post-hoc calibration. ROC AUC = 0.700 ± 0.054 indicates insufficient discrimination between WildChat benign and jailbreak conversations at the representation level. At TPR$\geq$95%, the optimal threshold still yields FPR = 62.7%—far above deployable levels. Temperature scaling cannot improve this trade-off, as ROC curves are invariant to monotonic score transforms. The AUC ceiling reflects genuine domain shift: WildChat benign and jailbreak representations overlap substantially, and no post-hoc calibration can overcome this; in-domain benign adaptation is required.

P Benign Distribution Diversification Pilot

To test whether the high WildChat FPR (Section O) reflects a fundamental limitation or a remediable distribution mismatch, we conduct a pilot retraining experiment mixing 150 WildChat benign conversations into the training set. We filter 27 of 226 WildChat conversations containing jailbreak-indicative patterns, use 150 for training and 49 for held-out evaluation. Both training and evaluation WildChat samples are drawn from the same source (GPT-3.5 users); this is therefore an IID evaluation of benign coverage, not an OOD generalization test.

Class ratio shift. Adding 150 WildChat benign to the original 350-conversation training set (175 jailbreak + 175 synthetic benign) shifts the benign-to-jailbreak ratio from 50:50 to approximately 65:35 (325 benign vs. 175 jailbreak). To control for this confound, we downsample WildChat benign to 88 conversations (175 jailbreak : 88 WildChat benign + 87 original benign $\approx$ 50:50, $n = 350$). This balanced variant achieves identical WildChat FPR (0%, same CI) while maintaining OOD detection (AA F1 = 0.991, FITD/DD F1 = 1.000), confirming that class ratio is not the driver of FPR reduction.

Table 20: Effect of benign distribution diversification on WildChat FPR and OOD jailbreak detection. Mixed training adds 150 WildChat benign to the original 350 synthetic conversations. 3-seed mean (seeds 42, 123, 456). WildChat evaluation is IID (same source as training WildChat).

Metric	Original	Mixed
WildChat FPR (%)	68.1	**0.0**
IID F1	1.000	1.000
DD OOD F1	1.000	1.000
AA OOD F1	$\sim$0.960	0.956
FITD OOD F1	1.000	1.000

Mixed benign training reduces WildChat FPR from 68.1% to 0% (exact binomial 95% CI upper bound: 6.1%, $n = 49$) with no degradation in cross-attack detection (ActorAttack F1 = 0.956, within 0.4 pp of original). This confirms the diagnosis from Section O: when representative benign data is available, the high FPR is remediable without sacrificing detection performance. The evaluation set is small ($n = 49$) and IID with respect to the training WildChat distribution, so the result should be interpreted as evidence that benign distribution mismatch—not model capacity—is the bottleneck; validation on larger and independently sourced OOD benign corpora is needed to establish deployment-ready FPR bounds.

Benign adaptation curve. To quantify how many OOD benign samples suffice, we vary the number of WildChat benign conversations added to training ($N \in \{0, 5, 10, 25, 50, 100, 150\}$) and retrain the full pipeline (9-class DAPT + BiGRU) for each N (seed 42).

Table 21: Benign adaptation curve: WildChat FPR and OOD detection F1 as a function of N WildChat benign conversations added to the 350-conversation training set. Seed 42 (single-seed curve); $N = 10$ confirmed across 3 seeds (FPR = 0% for all). WildChat evaluation $n = 49$.

N	Train	FPR (%)	DD F1	AA F1	FITD F1
0	350	65.3	1.000	0.981	1.000
5	355	16.3	0.990	0.981	1.000
10	360	**0.0**	1.000	1.000	1.000
25	375	0.0	1.000	0.953	1.000
50	400	0.0	1.000	1.000	1.000
100	450	0.0	1.000	1.000	1.000
150	500	0.0	1.000	0.981	1.000

FPR drops sharply from 65.3% ($N = 0$) to 16.3% ($N = 5$) to 0% ($N = 10$), remaining at 0% for all

$N \geq 10$. OOD detection is preserved throughout: all F1 scores ≥ 0.953 (minor ActorAttack variation at $N = 25$ is within seed noise). This demonstrates that benign distribution mismatch is addressable with minimal annotation effort—10 conversations from the target domain suffice to eliminate false positives entirely (95% CI upper bound: 6.1% at $n = 49$). IID F1 remains 1.000 for all N. Multi-seed validation at $N = 10$ (seeds 42, 123, 456) confirms robustness: FPR = 0.0% for all seeds, DD/FITD F1 = 1.000±0.000, AA F1 = 0.981±0.016.

Second benign source: OASST2. To test whether benign-side shift generalizes beyond WildChat, we evaluate on 200 benign multi-turn conversations from OpenAssistant OASST2 (Köpf et al., 2023)—a crowdsourced dataset of real human conversations, distinct from WildChat (GPT-3.5 user logs) in both authorship and topic distribution.

Table 22: OASST2 benign adaptation curve: FPR on 200 OASST2 benign conversations and OOD jailbreak detection F1 as a function of N OASST2 conversations added to training. $N = 25$ confirmed across 3 seeds (mean±std); other rows seed 42. Compare WildChat (Table 21): FPR reaches 0% at $N = 10$.

N	FPR (%)	IID F1	DD F1	AA F1	FITD F1
0	57.5	–	–	–	–
10	20.5	1.000	1.000	1.000	1.000
25	**1.0±0.5**	1.000	1.000	0.953±.037	1.000
50	**0.5**	1.000	1.000	0.990	1.000

Baseline FPR on OASST2 (57.5%) is comparable to WildChat (68.1%), confirming that benign-side shift is not source-specific but a general consequence of synthetic-only benign training. The adaptation curve follows the same pattern—monotonic FPR reduction with preserved OOD detection—but requires more samples: 25 conversations reduce FPR to 1.0%±0.5% across 3 seeds (vs. 0% at $N = 10$ for WildChat), and 50 conversations reach 0.5%. The slower convergence likely reflects greater distributional distance between crowdsourced OASST2 conversations and the synthetic training distribution compared to WildChat's GPT-3.5 user logs.

Q BERT-base Proxy Baseline

Table 24 compares BERT-base-uncased (110M parameters, matching DeepContext's encoder) against DeBERTa-v3-base (184M), both without domain-adaptive pretraining. The two vanilla encoders achieve comparable OOD performance (DD: 0.318 vs. 0.312; AA: 0.275 vs. 0.258), despite a 1.7× parameter difference and different pretraining procedures. Both collapse on OOD evaluation while 9-class DAPT recovers to 0.994 (DD) and 0.939 (AA). This confirms that cross-attack generalization is driven by domain adaptation, not encoder architecture or capacity at this scale.

Table 23: Benign-side adaptation summary across two independent sources. Both achieve deployment-level FPR (≤1.5%) with no degradation in OOD jailbreak detection. 3 seeds each.

Source	N	FPR (mean±std)	Seeds	Eval n
WildChat	10	0.0%±0.0%	3	49
OASST2	25	1.0%±0.5%	3	200

Table 24: BERT-base-uncased (110M) frozen probing without DAPT, matching DeepContext's encoder family (Albrethsen et al., 2026). 3-seed mean±std. DeBERTa-v3 results from Table 4 for comparison.

Encoder	IID	DD OOD	AA OOD
BERT-base (110M), no DAPT	.996±.006	.318±.031	.275±.027
DeBERTa-v3 (184M), no DAPT	1.000±.000	.312±.043	.258±.000
DeBERTa-v3 (184M), 9-class	1.000±.000	.994±.005	.939±.047

R Confidence Calibration

Table 25: Expected calibration error (ECE, 10 bins) for 9-class and vanilla on IID and OOD evaluation (3-seed mean±std). Lower is better.

Variant	IID	DD OOD	AA OOD
9-class	≈0.000	0.143±0.053	0.116±0.053
Vanilla	0.009±0.007	0.669±0.004	0.504±0.066

The 9-class model achieves substantially lower ECE than vanilla on OOD data (0.12–0.14 vs. 0.50–0.67), indicating that domain adaptation improves not only discriminative performance but also confidence calibration. IID calibration is near-perfect for both variants. The residual OOD ECE (0.12–0.14) suggests that temperature scaling could further improve deployment reliability.

S Frozen Probing Verification

Table 26 compares frozen probing across DAPT objectives and against end-to-end fine-tuning. On easy OOD (DD, FITD), frozen MLM and frozen 9-class representations perform comparably ($\Delta \leq 1.3$ pp). On hard OOD (ActorAttack), 9-class retains a modest advantage (0.972 vs. 0.941,

Table 26: Frozen probing vs. end-to-end (E2E) fine-tuning (3-seed mean±std). 9-class: plan_002 checkpoints; MLM: plan_017_mlm_control (matched training). Frozen probing trains only BiGRU+MLP on frozen DAPT embeddings.

Config	IID	DD OOD	AA OOD	FITD OOD
Frozen 9-class	1.000±.000	.997±.005	.972±.026	1.000±.000
Frozen MLM	.982±.016	.993±.009	.941±.037	.987±.009
E2E 9-class	1.000±.000	1.000±.000	.997±.005	—

Table 27: Representation geometry and class separability across Stage-1 variants. PR: participation ratio (effective dimensionality of the embedding space). Fisher ratio: inter-class distance^2 / sum of intra-class variances. Weight Δ: L2 distance from pretrained weights (relative to pretrained norm).

Variant	PR	Fisher	Wt. Δ (%)
Vanilla	31.9	<0.001	0.000
Wiki-MLM	25.0	<0.001	0.221
JB-MLM	24.6	<0.001	0.256
Topic	14.5	0.012	0.371
Scrambled	7.8	2.60	0.283
9-class	9.4	4.47	0.303
Binary	3.7	19.7	0.239

$\Delta = 3.1$ pp) with lower variance ($\sigma = 0.026$ vs. 0.037). Both far exceed vanilla encoders ($\leq$0.318; Table 24). The E2E advantage over frozen 9-class is $\leq$2.5 pp, confirming that Stage-1 representation quality is the primary source of OOD generalization.

A striking asymmetry emerges when comparing frozen probing to the full pipeline (Table 3): MLM frozen representations *outperform* the full MLM pipeline on AA OOD (0.941 vs. 0.893), while 9-class frozen exactly matches its full pipeline (0.972 vs. 0.972). This indicates that MLM's cross-seed instability originates in Stage-2 BiGRU training rather than Stage-1 DAPT—the MLM encoder produces adequate representations that are then degraded by unstable downstream optimization. Classification DAPT, by contrast, produces representations that remain robust through Stage-2 training, explaining its consistently lower variance (Table 3).

This two-level structure—DAPT vs. no-DAPT producing a representation quality leap (+70 pp), classification vs. MLM DAPT producing a stability difference—is consistent with two further observations: λ=0 matches $\lambda > 0$ during E2E fine-tuning (Table 34), and mean pooling matches BiGRU under E2E (Table 29). Once DAPT representations are strong, downstream configuration details have minimal impact.

Representation geometry analysis. To characterize the mechanism behind classification DAPT's robustness, we analyze Stage-1 representations across all seven DeBERTa-v3-base variants (seed 42). We extract `[CLS]` embeddings from 150 test conversations (50 IID + 50 DD OOD + 50 AA OOD) and compute three metrics.

Classification variants compress representations into 2–9 effective dimensions (vs. 25–32 for MLM/vanilla) while increasing Fisher ratio by 3–5 orders of magnitude. Weight distances from pretrained are comparable across all variants (∼0.2–0.4%), indicating that the mechanism is representational restructuring rather than parameter-space regularization. Scrambled labels produce similar compression (PR = 7.8, Fisher = 2.6), confirming that the classification objective structure—not label semantics—drives discriminative compression.

Layer-wise localization. To determine *where* discriminative compression occurs, we compute PR and Fisher ratio at every DeBERTa-v3-base layer (0–12) for four variants (9-class, scrambled, JB-MLM, vanilla) using the same 236-sample evaluation set (seed 42).

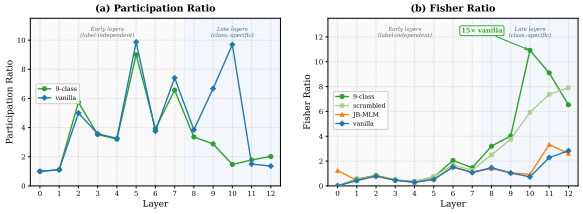


Figure 6: Layer-wise representation analysis. (a) Participation ratio: classification variants (9-class, scrambled) show sharp dimensional collapse at layers 9–10 (PR ≈ 1.5–2.9) while vanilla/JB-MLM maintain high dimensionality (PR ≈ 7–10). (b) Fisher ratio (log scale): class separability diverges from layer 8, with 9-class peaking at 10.9 (layer 10) vs. vanilla 0.73 (15×).

Layers 0–7 show no systematic divergence across variants (Fisher < 2.1 for all; PR tracks similar oscillation patterns), consistent with Maennel et al. (2020)'s finding that early layers learn label-independent features. From layer 8 onward, classification-supervised variants diverge sharply: 9-class Fisher rises from 3.2 (layer 8) to 10.9 (layer 10, peak), while vanilla remains below 1.5 throughout. Simultaneously, 9-class PR drops from 3.4 to 1.5 at layer 10—a 6.6× compression relative to vanilla (PR = 9.7). JB-MLM tracks vanilla across

all layers (Fisher ≤ 3.3, PR ≥ 1.4), confirming that MLM does not induce discriminative restructuring. Scrambled labels exhibit the same late-layer pattern (Fisher peaks at 7.9, layer 12; PR drops to 1.5, layer 12), reinforcing that any classification signal—regardless of label semantics—triggers late-layer compression.

Causal subspace probe. To test whether the compressed subspace is *sufficient* for OOD detection, we project layer-10 `[CLS]` representations onto the top-k SVD components and train a logistic regression probe (3 seeds, sklearn default). Controls: random k-dimensional projections (10 random bases, mean reported) and vanilla/MLM models under identical projection.

Table 28: Causal subspace probe: OOD F1 when projecting layer-10 representations onto top-k SVD components vs. random k-dimensional projections. 9-class top-3 achieves perfect AA detection with 3/768 dimensions; MLM fails even with all 768.

Model	Projection	DD	AA	FITD
9-class	top-3	1.000	**1.000**	1.000
9-class	top-5	1.000	1.000	1.000
9-class	random-3	1.000	0.928±.129	1.000
9-class	random-5	1.000	0.952	1.000
9-class	full-768	1.000	1.000	1.000
scrambled	top-3	1.000	0.962	1.000
scrambled	random-3	0.981	0.766	0.987
JB-MLM	top-3	1.000	0.618	1.000
JB-MLM	full-768	1.000	0.636	1.000
JB-MLM	random-5	0.685	0.584	0.744

Three findings establish a causal link between compression and generalization (Table 28). First, 9-class top-3 achieves perfect ActorAttack detection (F1 = 1.000) using only 3 of 768 dimensions—all discriminative information resides in the compressed subspace. Second, random 3-dimensional projections degrade to 0.928 on AA, confirming that the specific SVD directions matter. Third, JB-MLM reaches only 0.636 on AA even with all 768 dimensions, demonstrating that without classification-induced compression, the representation lacks the structure needed for cross-attack generalization. DD and FITD are near-ceiling for all models, making ActorAttack the critical test.

T Temporal Aggregation Comparison

Table 29 compares BiGRU and mean pooling under both frozen and E2E settings. Under frozen probing (the main paper's pipeline), BiGRU provides a marginal advantage on ActorAttack (+1.0 pp) with overlapping confidence intervals, consistent with the main text finding that temporal order is not a core detection factor ($|\Delta| < 1$ pp under turn shuffling; Section 4.3). Under E2E fine-tuning, the difference vanishes entirely: mean pooling saturates all benchmarks at F1 = 1.000. The choice of frozen BiGRU in the main pipeline reflects a design trade-off: it enables representation-level analysis (CKA, t-SNE) that E2E would confound, while sacrificing ≤ 1 pp on the hardest OOD set.

Table 29: Temporal aggregation comparison: mean pooling vs. BiGRU under frozen and end-to-end (E2E) settings (3-seed mean±std). Plan_002 9-class checkpoints.

Setting	Aggregator	DD OOD	AA OOD
Frozen	BiGRU	.997±.005	.972±.026
	Mean pooling	.991±.013	.962±.015
E2E	BiGRU	1.000±.000	.997±.005
	Mean pooling	1.000±.000	1.000±.000

U Length Stratification Analysis

Table 30: Jailbreak recall by conversation length bin on OOD evaluation (3-seed majority vote). Benign conversations span 12–20 turns; jailbreak conversations span 3–10 turns.

Dataset	Variant	1–3 turns	4–6 turns	7–10 turns
DD OOD	9-class	—	100.0%	99.4%
	Vanilla	—	$\leq$2.6%	$\leq$1.7%
AA OOD	9-class	93.3%	95.6%	100.0%
	Vanilla	0.0%	$\leq$2.6%	0.0%

Table 30 stratifies jailbreak recall by conversation length. The 9-class model shows no systematic degradation: ActorAttack recall increases monotonically from 93.3% (1–3 turns) to 100% (7–10 turns), and DD recall is near-perfect across all bins.

Length confound. Benign conversations (12–20 turns) and jailbreak conversations (3–10 turns) have zero length overlap in the OOD evaluation sets, raising the concern that the model discriminates by conversation length rather than content. Three observations argue against this interpretation: (1) Vanilla DeBERTa has access to identical length information via the BiGRU but achieves recall $\leq$2.6% across all bins—length alone is insufficient without domain-adapted representations. (2) At prefix K=1, where all conversations are length-equalized,

9-class still achieves F1 = 0.773±0.041 on ActorAttack (Table 9), demonstrating content-based detection independent of length. (3) The model is trained on IID data (Crescendo) and cannot learn OOD-specific length distributions; generalization to unseen length patterns must rely on content features that transfer across attack families.

Table 31: Turn-count independence verification (9-class, plan_002 checkpoints, 3 seeds). Part A: FPR on test benign truncated to K user turns (all seeds FPR = 0). Part B: WildChat FPR by turn-count bucket (mean±std). Part C: jailbreak detection (F1) after padding with benign turns to 15+ total turns (mean±std where available).

Condition	FPR / F1	n
Part A: Benign truncation (test set)		
Original (12–20 turns)	FPR = .000	38
Truncated K=3	FPR = .000	38
Truncated K=5	FPR = .000	38
Truncated K=8	FPR = .000	38
Truncated K=10*	FPR = .000	38
Part B: WildChat FPR by turn count		
Short (3–10 turns)	FPR = .770±.105	196
Medium (11–15 turns)	FPR = .942±.054	23
Long (16+ turns)	FPR = .857±.117	7
Part C: Jailbreak padding to 15+ turns		
DD original	F1 = .996	80
DD padded	F1 = .971 (Δ=−2.5 pp)	80
AA original	F1 = .963	80
AA padded	F1 = .383±.261 (Δ=−57.9 pp)	80

* K=10 did not truncate any conversation (test benign max ∼8 user turns).

Direct empirical verification. Truncating benign conversations to match jailbreak turn counts (Part A) produces zero false positives across all truncation levels and all 3 seeds, directly confirming that the model does not use conversation length as a detection shortcut. K=3 and K=5 are the effective control conditions where all 38 conversations were truncated; K=10 exceeds the maximum test benign length and serves as a no-op control.

WildChat FPR by turn count (Part B) provides complementary evidence: short conversations (3–10 turns, n=196) have *lower* FPR than medium-length ones (11–15 turns, n=23), inconsistent with a "short = jailbreak" shortcut. Medium and long buckets have limited sample sizes and should be interpreted as directional evidence only.

Padding jailbreak conversations with benign turns (Part C) reveals an asymmetry: DD detection is resilient (Δ=−2.5 pp), but ActorAttack collapses (Δ=−57.9 pp). Padding experiments append benign turns *after* the complete attack sequence; in practice, adversaries would not dilute their own successful attacks. The collapse nonetheless exposes BiGRU temporal aggregation as vulnerable to attention dilution on subtle attack signals, a limitation discussed in Section 7.

V Efficiency

Table 32: Inference efficiency comparison on 100 conversations (50 IID + 50 DD OOD), same GPU (RTX PRO 6000). All methods measured under identical conditions.†

Method	Latency (ms)	Params	VRAM
DeBERTa+BiGRU (ours)	13.1±1.6	186.7M	868 MB
Qwen3-8B 3-shot	318.5±26.8	8.2B	16.1 GB

† LLM inference speed varies significantly across GPU generations; same-hardware comparison ensures a fair speedup estimate.

Table 32 compares inference cost under identical conditions. DeBERTa+BiGRU processes a conversation in 13.1 ms, 24.3× faster than Qwen3-8B 3-shot (318.5 ms) while achieving substantially higher OOD detection (DD = 0.566, AA = 0.652 for Qwen3-8B; Table 6). The encoder pipeline requires 868 MB GPU memory (18.5× less than the LLM), enabling deployment on consumer hardware.

V.1 Data Efficiency

Table 33: Data efficiency of 9-class DAPT (3-seed mean±std, sample std with $n-1$ divisor) and vanilla baseline (seed 42 only). N = total training conversations (balanced jailbreak/benign).

N	Variant	IID	DD OOD	AA OOD
50	9-class	1.000±.000	1.000±.000	.969±.037
125	9-class	1.000±.000	1.000±.000	.975±.019
250	9-class	1.000±.000	1.000±.000	.975±.035
350	9-class	1.000±.000	.978±.030	.972±.033
50	Vanilla	.646	.529	.234
125	Vanilla	.532	.244	.244
250	Vanilla	.987	.365	.502
350	Vanilla	1.000	.360	.489

Table 33 shows that 9-class DAPT reaches near-saturation performance with as few as 50 conversations (25 jailbreak + 25 benign): AA OOD F1 = 0.969±0.037, already surpassing vanilla at full training size (N=350, AA = 0.489) by 48 pp using only 14% of the data. DD OOD is perfectly solved at all training sizes. Vanilla performance is noisy across N (single seed); the consistent pattern is that vanilla never exceeds DD = 0.53 or

AA = 0.50 regardless of training size, while 9-class maintains AA $\geq$ 0.969 at all N. Seed 456 is systematically lower across all N values, contributing most of the cross-seed variance; this is consistent with the initialization sensitivity observed in the main results.

W Auxiliary Loss Weight Ablation

Table 34: Effect of auxiliary persuasion loss weight λ during Stage-2 E2E fine-tuning (seed 42). Stage-1 DAPT encoder is identical across all rows; only the Stage-2 loss $L = L_{\text{binary}} + \lambda \cdot L_{\text{strategy}}$ varies.

λ	IID	DD OOD	FITD OOD	AA OOD
0	1.000	1.000	1.000	1.000
0.1	1.000	1.000	1.000	.990
0.5	1.000	1.000	1.000	.981
1.0	1.000	1.000	1.000	.981
2.0	1.000	1.000	1.000	.944

During Stage-2 E2E fine-tuning, the auxiliary persuasion loss weight λ has limited impact on detection (Table 34): the maximum AA OOD gap across all λ values is 5.6 pp (single seed), comparable to the cross-seed variance ($\sim$6 pp) observed in the main results. All configurations converge to perfect validation F1 within epoch 1, indicating that DAPT representations already encode sufficient discriminative features for downstream classification (Section S). The value of persuasion strategy classification lies in Stage-1 DAPT—where it shapes encoder representations that transfer across attack families—not in the Stage-2 auxiliary loss.

X OOD Strategy Prediction Analysis

Table 35: OOD strategy prediction accuracy (jailbreak turns only) and detection rate for the 9-class model (3-seed mean). r: Pearson correlation between per-strategy prediction accuracy and per-strategy detection rate.

Dataset	Strat. Acc.	Det. Rate	Missed (per seed)	r
DD OOD	50.1%	99.6%	0 / 0 / 1	$\sim$0.45
AA OOD	50.1%	96.2%	1 / 7 / 1	$\sim$0.24

Table 35 reports OOD strategy prediction accuracy alongside detection rates. Strategy prediction accuracy is substantially above random (50.1% vs. 11.1% chance for 9 classes) and positively correlated with detection success ($r \approx 0.24$–0.45), but strategy prediction is not the detection mechanism: frozen probing with a randomly initialized strategy head (Appendix S) achieves 97.2% AA OOD detection, and λ=0 (no strategy loss) matches or exceeds $\lambda > 0$ during E2E fine-tuning (Table 34). The classification objective shapes representations during Stage-1 DAPT training; at inference, strategy predictions are a byproduct of those representations, not a driver of detection.

Y Cross-Seed Ensemble

Table 36: Majority vote ensemble (3 seeds) vs. single-seed mean on OOD Full. "Fixed" = samples corrected by ensemble; "Broke" = samples corrupted by ensemble.

Variant	DD single→ens.	AA single→ens.	Fixed	Broke
9-class	.997→**1.000**	.972→**.981**	5	0
MLM	.914→**1.000**	.897→**.953**	26	0
Vanilla	.258→.258	.258→.258	0	0

Table 36 shows that majority vote never corrupts correct predictions (Broke = 0 for all variants). MLM benefits most from ensembling (+8.6 pp DD, +6.0 pp AA), because its per-seed errors are complementary—each seed fails on different samples, and the majority corrects these. This further exposes MLM's seed instability: its individual seeds are unreliable, but their errors are uncorrelated. By contrast, 9-class seeds are already highly consistent (std $\leq$ 0.026), leaving minimal room for ensemble improvement (+0.3–0.9 pp). Vanilla errors are systematic (all seeds fail on the same OOD samples), so ensembling provides no benefit.

Z Seed-Level F1 Comparisons

To verify that cross-seed pooling does not mask inconsistent pairwise orderings, Table 37 reports per-seed F1 for all variants on OOD DD and ActorAttack at full conversation length. The relative ordering of variants is consistent across all three seeds: classification-supervised variants (9-class, binary, scrambled) dominate MLM and vanilla at every seed, and no seed reverses the hierarchy established by the pooled analysis. Table 38 extends the comparison between 9-class and scrambled to 5 seeds for TOST analysis.

Cross-hardware replication. To assess hardware sensitivity, we independently retrained 9-class and JB-MLM from scratch on a different GPU (RTX PRO 6000 Blackwell, 5 seeds each). Table 39 reports OOD F1 on all three attack families. IID and FITD results are perfectly consis-

Table 37: Per-seed OOD F1 (Full) for all variants across three attack families (seeds 42, 123, 456). Cohen's d for 9-class vs. scrambled: DD = -0.003 (negligible), AA = 0.587 (medium; CI crosses 0 at n=3).

	DD			ActorAttack			FITD		
Variant	s42	s123	s456	s42	s123	s456	s42	s123	s456
9-class	1.000	1.000	.990	.990	.935	.990	1.000	1.000	1.000
Binary	1.000	1.000	1.000	.981	.990	1.000	1.000	1.000	1.000
Scrambled	.990	1.000	1.000	.891	.990	.971	1.000	1.000	1.000
MLM+cls	1.000	.990	.751	.971	.925	.784	.961	.432	.729
Topic	.727	.252	.644	.686	.431	.759	.990	.774	.909
Vanilla	.284	.305	.348	.258	.258	.258	.871	.855	.919

Benign recall = 1.000 for all DAPT variants across all seeds and OOD conditions (IID benign); OOD F1 variation is driven entirely by attack recall. Classification-supervised variants (9-class, binary, scrambled) achieve perfect or near-perfect F1 on FITD, confirming its non-discriminative nature.

Table 38: 5-seed OOD F1 (Full) for 9-class vs. scrambled (seeds 42, 123, 456, 789, 1000). At $n = 10$ (5 additional seeds; see text), TOST (δ=5 pp): DD $p < 10^{-8}$ ✓; FITD $p < 10^{-8}$ ✓; AA $p = 0.183$ (non-equivalent). Cohen's d at $n = 10$: DD ≈ 0; AA = 0.44 (medium).

Attack	**Variant**	s42	s123	s456	s789	s1000	Mean±SD
DD	9-class	1.000	1.000	.990	1.000	1.000	.998±.004
DD	Scrambled	.990	1.000	1.000	1.000	1.000	.998±.004
AA	9-class	.990	.935	.990	.972	.972	.972±.021
AA	Scrambled	.891	.990	.971	.990	.990	.966±.040

tent (9-class: 5/5 seeds = 1.000 on both). On ActorAttack, 9-class outperforms MLM directionally ($\Delta = +1.7$ pp) but the difference is not statistically significant (paired t-test $p = 0.246$), reflecting the higher variance inherent to this structurally complex attack family; this is consistent with the complexity-dependent boundary reported in Section 4.3.

Table 39: Cross-hardware 5-seed replication (RTX PRO 6000 Blackwell). OOD F1 (Full) for 9-class vs. JB-MLM across three attack families.

Attack	**Variant**	s42	s123	s456	s789	s1000	Mean±SD
IID	9-class	1.000	1.000	1.000	1.000	1.000	1.000±.000
IID	JB-MLM	1.000	1.000	1.000	1.000	1.000	1.000±.000
DD	9-class	1.000	1.000	1.000	1.000	1.000	1.000±.000
DD	JB-MLM	1.000	1.000	1.000	1.000	1.000	1.000±.000
FITD	9-class	1.000	1.000	1.000	1.000	1.000	1.000±.000
FITD	JB-MLM	1.000	1.000	1.000	1.000	1.000	1.000±.000
AA	9-class	.990	.943	.972	.972	.972	.970±.016
AA	JB-MLM	.857	.751	.906	.916	.943	.875±.074

AA difference is directionally consistent ($\Delta = +9.5$ pp mean) but reflects high MLM variance; paired t-test $p = 0.246$. Template attacks (DD, FITD) show perfect agreement across all seeds and both variants.